VECTOR

3rdParty HSM_MEM Integration

Technical Reference

Infineon TC3xx

Version 1.0.0

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Status	Released

Document Information

History

Author	Date	Version	Remarks
Stefan Schmid	2020-07-17	1.0.0	Basic Integration of TC3xx HsmMem; Integration of HSM_MEM_TC3xx_1.0.0
Roland Suess	2020-08-07		Adaptation for usage of HsmMem instead of MCAL package

Reference Documents

No.	Source	Title	Version
[1]	Vector	TechnicalReference_3rdParty-HsmMem-Integration_TC3xx.pdf	See Delivery
[2]	Vector	ScreenCast_McalIntegration_Tresos.pdf	See Delivery

Scope of the Document

This document contains information about the integration of 3rd Party HSM_MEM into Vector software stack.

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1 HSM_MEM Integration

1.1 Type of Integration

Basic Integration
Both configuration tools, EB tresos™ as well as Vector DaVinci Configurator, are used for configuration.

Recommended workflow:
Start initial configuration with EB tresos™, export it in AUTOSAR format and import it into Vector DaVinci Configurator. Generation and minor changes in configuration are done in Vector DaVinci Configurator.
For usage with Vector DaVinci Configurator 5 please refer to chapter 'Mixed configuration tool usage' in the document `TechnicalReference_3rdParty-HsmMem-Integration.pdf` [1].



Multimedia Link
For additional information please also refer to the Vector screen cast referenced in ScreenCast_McalIntegration_Tresos.pdf [2]

1.2 HSM_MEM Location within SIP

The 3rd Party Software is separated from the Vector parts within the SIP. It might not even be part of the delivery. Please refer to chapter 'First Steps' in document `TechnicalReference_3rdParty-HsmMem-Integration.pdf` [1].

1.3 Supported 3rd Party Products

This integration supports the following Infineon packages:

- HSM_MEM_TC3xx_1.0.0
- HSM_MEM_TC3xx_1.0.0-rc

Developed for the targets:

- TC39x B-Step
- TC38x A-Step
- TC37x, TC37x_ED A-Step
- TC36x A-Step
- TC35x A-Step
- TC33x, TC33x_ED A-Step

- TC32x A-Step



Note
Please refer to the Release Notes of the 3rd Party Products for further information, e.g. regarding supported versions, derivatives and compilers.



Note

Only official 3rd Party vendor releases are part of this Vector integration package. Therefore, any customer-specific releases cannot be considered.



Caution
To find out if there are further Hotfixes available for your HsmMem package, please contact the 3rdParty Vendor.

It is essential to replace the affected EB tresos™ module plugins in your original package before you start 3rdPartyMcalIntegrationHelper.exe.

1.4 Configuration Tools

- Vector DaVinci Configurator 5
- EB tresos™

2 Vector Comment

The user should consider the attached `TechnicalReference_3rdParty-MCAL-Integration.pdf` [1] for further information regarding Vector integration and setup of a project.

2.1 Known Issues

The HsmMem package in use might not be the latest one. Updates or Hot Fixes might be available from the vendor.

If you discover errors in the HsmMem during development or suspect that faulty behavior could be caused by the HsmMem, please contact the 3rdParty Vendor. If there are updates for your HsmMem available, it might be helpful to use them. The corresponding 3rdParty Vendor Release Notes will tell you which errors have been fixed.

The errors documented in the following chapters were detected at Vector during the HsmMem integration and reported to the 3rdParty Vendor.



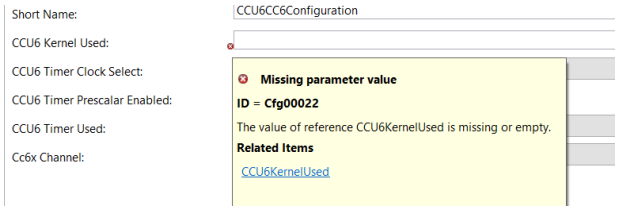
Note
Vector makes every effort to ensure that this integration is compatible with all HsmMem packages of the corresponding Controller Family as long as the 3rdParty Vendor does not change the structure of the packages. However, not all HsmMem packages can be tested and therefore the user may find errors either in Vector's files or in the HsmMem that are not listed in this document.



Caution
Modifications to the HsmMem, which may be described in the following chapters under the keyword 'Workaround', must take place after the 3rdPartyMCALIntegrationHelperTool has run. Otherwise the changes will be overwritten.

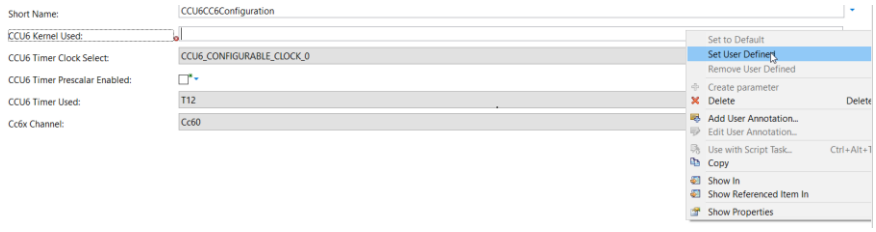
2.1.1 General Issues

2.1.1.1 Inconsistency between Tresos and CFG5



There may be inconsistencies between parameters in Tresos and CFG5. This is because the parameter in Tresos is marked as optional while it is mandatory in CFG5.

To avoid that the parameter has to be configured in CFG5 and the configuration is changed, set the parameter to User Defined.



2.1.2 All derivates

2.1.2.1 HSM_MEM_TC3xx_1.0.0-rc / HSM_MEM_TC3xx_1.0.0

2.1.2.1.1 IfxHsm.h: No such file or directory

The file Hsm_TimSer.c includes the header file IfxHsm.h which is not part of the Hsm package.



Workaround

Add a stub file IfxHsm.h in the folder “Appl/include” with the following content:

#include "IFX_HSM.h"

2.2 Restrictions

N/A

3 Glossary and Abbreviations

3.1 Glossary

Term	Description
3rd party components / MCAL	BSW modules not provided by Vector. Vector may have integrated the software within the SIP but does not take over any responsibility regarding functionality of these modules.
DaVinci Configurator	Configuration and generation tool for Vector MICROSAR components

Table 3-1 Glossary

3.2 Abbreviations

Abbreviation	Description
MCAL	Microcontroller Abstraction Layer
AUTOSAR	Automotive Open System Architecture
SIP	Software Integration Package (as provided by Vector)
Msn	Module Short Name according AUTOSAR

Table 3-2 Abbreviations

4 Contact

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